



# Examiners' Report Summer 2026

Pearson Edexcel GCE

In AS Further Mathematics (8FM0)

Paper 26 Further Mechanics 2

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## General Comments

This paper allowed learners to demonstrate a wide range of Further Mechanics knowledge and skills, with many responses showing secure understanding of standard techniques and modelling approaches. The most successful learners combined accurate mathematical work with clear reasoning and carefully structured solutions.

A recurring theme throughout the paper was that learners often made good progress when they selected efficient methods and worked with exact values. Where marks were lost, this was frequently due to incomplete justification, algebraic slips, omission of relevant information from a model, or the use of less efficient approaches that increased the likelihood of error. Centres should continue to encourage learners to communicate their reasoning clearly, make effective use of diagrams, and check that all aspects of a situation have been incorporated into their mathematical model.

## Comments on Individual Questions

### Question 1

Responses to this question showed strong familiarity with the relevant concepts and methods. Stronger responses formed moments equations for all parts. Some responses would have been more successful if they had made equations instead of just expressions; in particular many responses could have scored marks in part (a) if they had included " $\bar{x} =$ ". Part (c) required forming a new equation and a number of responses did not include the mass of the fourth particle in the total mass.

### Question 2

The first part of this question provided an accessible start, with many correct and efficient responses. Some of the responses would have been improved by using a geometrical approach rather than a forces approach.

A number of responses made good progress using less efficient methods, such as the cosine rule. Some responses did not gain all the marks because they worked inexactly, for example by using angles of  $36.9^\circ$  or similar.

In part (b), most responses formed appropriate equations, usually by resolving vertically and horizontally. Some responses resolved along  $AP$  and  $BP$ ; these were more successful when the acceleration was treated correctly. Responses that did not reach the final answer often showed errors in algebraic manipulation.

The final part of the question was the most challenging aspect. Some responses considered only the tension in  $AP$ ; to gain the relevant marks, responses also needed to consider the tension in  $BP$ .

### Question 3

Part (a) of this question proved to be challenging. Many responses did not gain marks as they used  $\frac{\pi}{2}$  instead of  $\frac{\pi}{4}$ . Some responses which used the correct angle would have scored both marks if they had included a clear explanation or mark on a diagram rather than just multiplying their expression by  $\frac{1}{\sqrt{2}}$  without explanation.

The second part of this question was well answered, but some responses did not observe that the densities were different. Many responses to part (c) were more successful; other responses would have gained more marks if they had been able to correctly identify suitable lengths for a trigonometric ratio.

#### Question 4

Most responses to part (a) were good and integrated into something of the form  $-2 \ln\left(3 - \frac{1}{2}v\right)$  or  $-2 \ln(6 - v)$  but some responses would have been more successful if they had avoided sign errors, particularly when integrating. Almost all responses which attempted part (b) were correct. Perhaps because of the instruction, most responses to part (c) used integration and only a small number of *suvat*-style attempts were seen, which did not score any marks. More responses would have been successful if they had recognised that the values of 4 and 5 related to velocity and therefore needed to be converted to corresponding values of time. A small number of strong responses opted to form a differential equation in  $v \frac{dv}{dx}$  as noted in the alternative mark scheme.